Hardware-Level Adaptive Machine Learning For Embedded & IoT Systems

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INTRODUCTION

Machine learning deployed on embedded and IoT devices often referred to as TinyML enables low-latency, energy-efficient inference close to the data source. Despite these benefits, most TinyML models rely on static, pre-deployment optimizations such as quantisation, pruning, and model compression.

However, embedded devices operate under highly dynamic conditions, including:

- **Thermal drift**
- **Battery voltage drop**
- **Component aging**
- **Memory pressure**
- **Intermittent power (energy-harvesting IoT systems)**

These changing conditions cause accuracy degradation, latency spikes, and increased energy consumption, yet current models do not adapt to hardware changes during inference.

PROBLEM STATEMENT

There is no lightweight runtime adaptation framework that allows embedded ML models to adjust to hardware variability.

Existing research focuses mainly on static optimization, leaving a gap in:

- **Real-time hardware monitoring**
- **Dynamic model-switching**
- **Mixed-precision adaptation**
- **Aging-aware inference**
- **Energy-aware runtime control**

This poster presents a research proposal to address this gap.

RESEARCH QUESTION, LITERATURE & METHOD

Research Question

How can lightweight, runtime hardware-level adaptation improve the robustness and energy efficiency of machine learning models on embedded and IoT devices operating under dynamic hardware constraints?

Hypothesis

If embedded ML models incorporate real-time hardware feedback loops to adjust inference behaviour dynamically, then they will achieve improved accuracy stability, energy efficiency, and performance robustness.

Key Literature Insights

Recent studies show:

✓ TinyML Maturity

Surveys (Heydari et al., 2025; Tsoukas, 2024) highlight strong maturity in model compression and hardware-aware model design.

✓ Quantisation Calibration

QCore (Campos et al., 2024) demonstrates small on device calibration improvements.

✓ Energy-Aware TinyML

Sabovica et al. (2025) show model selection based on available energy in self-powered devices.

✓ Hardware Aging Effects

Lanzieri (2024) documents temperature & aging-induced embedded hardware degradation.

❗ BUT:

No work integrates all hardware signals (temp, voltage, memory) into a runtime ML adaptation engine on constrained devices. This is the research gap.

Proposed Methodology

1. Hardware Profiling Module

- Collect runtime signals:
- Voltage
- Temperature
- Memory availability
- CPU load
- Duty cycle

2. Adaptation Engine (Artefact)

Lightweight controller triggers actions such as:

- Model variant switching (INT8 vs Pruned)
- Mixed-precision inference
- Early-exit / layer skipping
- Micro-calibration adjustments

3. Hardware Platforms

- Cortex-M4 MCU
- ESP32 SoC
- Raspberry Pi Zero / Coral Edge TPU

4. Evaluation

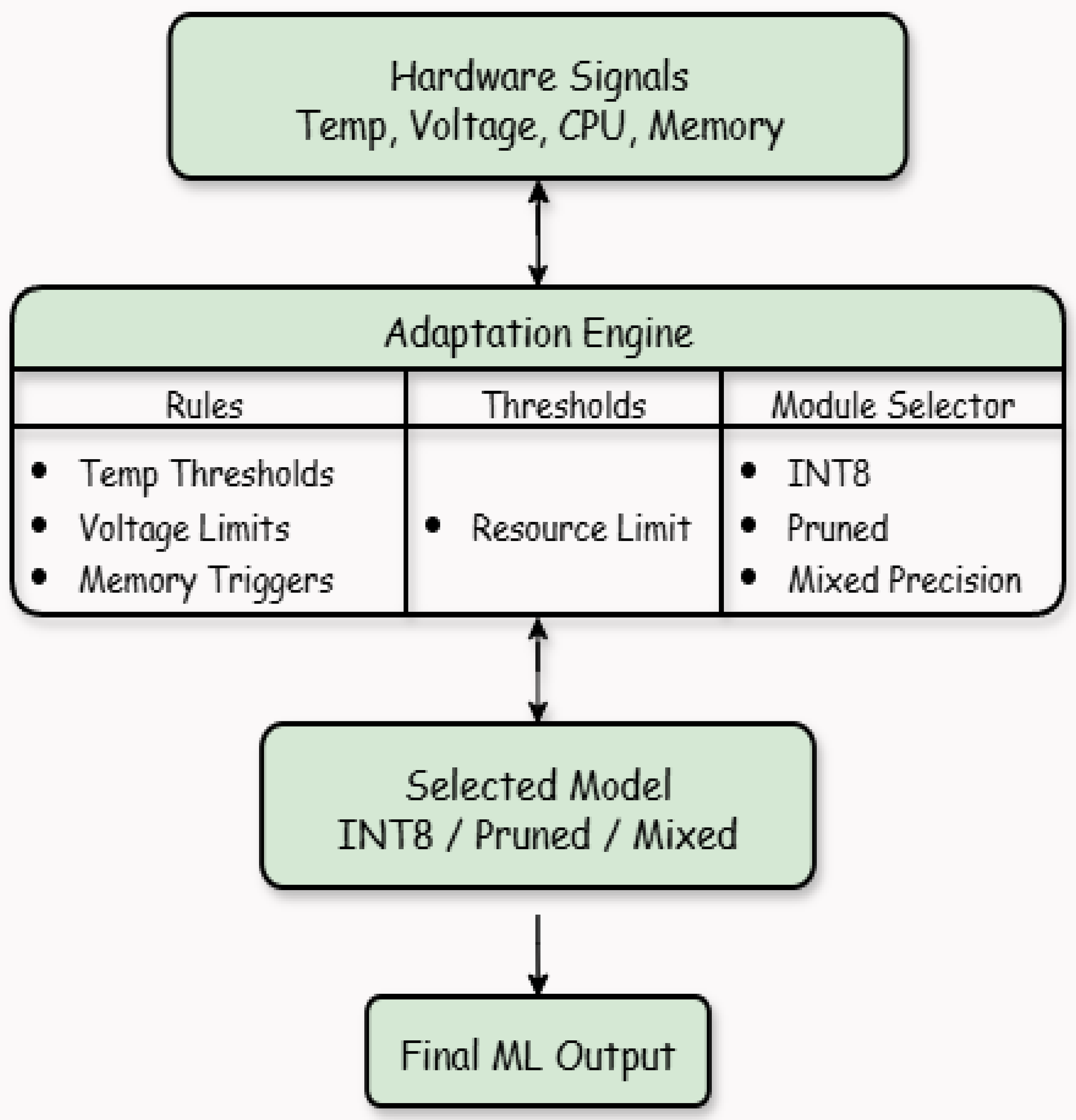
Test performance under simulated:

- Thermal drift
- Power fluctuation
- Aging-induced slowdown
- Intermittent energy availability

Metrics:

- Accuracy
- Latency
- Energy per inference
- Adaptation overhead

FULL SYSTEM BLOCK DIAGRAM



CONTRIBUTIONS & FUTURE RESEARCH

Expected Contributions

- First lightweight hardware-adaptive ML framework for TinyML devices.
- Improved energy efficiency and inference robustness.
- New insights into combining hardware drift and model behaviour.
- Practical artefact deployable on real embedded boards
- Reproducible Evaluation Dataset.

Future Research Directions

- Reinforcement-learning-based adaptation policies.
- Security-aware adaptation (avoid adversarial triggers).
- Ultra-low-power FPGA-based adaptation controllers.
- On-device lifelong learning for embedded systems.

REFERENCES

- Heydari, M., Faraji, S., & Rahmani, A. (2025). A survey on TinyML techniques and deployment challenges. Journal of Embedded Intelligence, 12(1), 45–72.
- Tsoukas, K. (2024). TinyML: Emerging methods for ultra-low-power ML deployment. ACM Computing Surveys, 56(4), 1–29.
- Campos, R., Alvarez, P., & Silva, J. (2024). QCore: Lightweight quantization calibration for embedded inference. IEEE Transactions on Neural Systems, 35(2), 110–123.
- Sabovica, T., Malik, U., & Petrov, D. (2025). Energy adaptive model selection for battery less IoT devices. Proceedings of the International Conference on Low-Power AI Systems, 88–96.